Q1 - 24 January - Shift 1

A 100 m long wire having cross-sectional area $6.25 \times 10^{-4}\ m^2$ and Young's modulus is $10^{10}\ Nm^{-2}$ is subjected to a load of 250 N, then the elongation in the wire will be :

(1) 6.25×10^{-3} m (2) 4×10^{-4} m
(3) 6.25×10^{-6} m (4) 4×10^{-3} m

Space for your notes:

Q2 - 24 January - Shift 2

Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A: Steel is used in the construction of buildings and bridges.

Reason R: Steel is more elastic and its elastic limit is high.

In the light of above statements, choose the most appropriate answer from the options given below

(1) Both **A** and **R** are correct but **R** is **NOT** the correct explanation of **A**

(2) **A** is not correct but **R** is correct

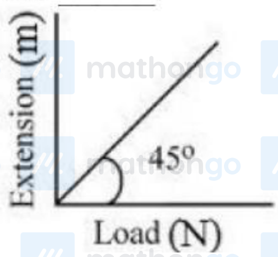
(3) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(4) **A** is correct but **R** is not correct

Space for your notes:

Q3 - 25 January - Shift 1

As shown in the figure, in an experiment to determine Young's modulus of a wire, the extension-load curve is plotted. The curve is a straight line passing through the origin and makes an angle of $45°$ with the load axis. The length of wire is 62.8 cm and its diameter is 4 mm. The Young's modulus is found to be $x \times 10^4$ Nm^{-2}.
The value of x is ______.



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Q4 - 30 January - Shift 1

Choose the correct relationship between Poisson ratio (σ). bulk modulus (K) and modulus of rigidity (η) of a given solid object:

(1) $\sigma = \dfrac{3K - 2\eta}{6K + 2\eta}$ (2) $\sigma = \dfrac{6K + 2\eta}{3K - 2\eta}$

(3) $\sigma = \dfrac{3K + 2\eta}{6K + 2\eta}$ (4) $\sigma = \dfrac{6K - 2\eta}{3K - 2\eta}$

Space for your notes:

Q5 - 30 January - Shift 2

A force is applied to a steel wire 'A', rigidly clamped at one end. As a result elongation in the wire is 0.2 mm. If same force is applied to another steel wire 'B' of double the length and a diameter 2.4 times that of the wire 'A', the elongation in the wire 'B' will be (wires having uniform circular cross sections)

(1) 6.06×10^{-2} mm
(2) 2.77×10^{-2} mm
(3) 3.0×10^{-2} mm
(4) 6.9×10^{-2} mm

Space for your notes:

Q6 - 31 January - Shift 1

A thin rod having a length of 1 m and area of cross-section $3\times10^{-6}\,m^2$ is suspended vertically from one end. The rod is cooled from 210°C to 160°C. After cooling, a mass M is attached at the lower end of the rod such that the length of rod again becomes 1 m. Young's modulus and coefficient of linear expansion of the rod are $2\times10^{11}\,N\,m^{-2}$ and $2\times10^{-5}\,K^{-1}$, respectively. The value of M is______ kg. (Take $g = 10\,m\,s^{-2}$)

Space for your notes:

Q7 - 31 January - Shift 2

For a solid rod, the Young's modulus of elasticity is 3.2×10^{11} Nm^{-2} and density is 8×10^3 kg m^{-3}. The velocity of longitudinal wave in the rod will be.

(1) 145.75×10^3 ms^{-1}
(2) 3.65×10^3 ms^{-1}
(3) 18.96×10^3 ms^{-1}
(4) 6.32×10^3 ms^{-1}

Space for your notes:

Q8 - 31 January - Shift 2

Under the same load, wire A having length 5.0 m and cross section 2.5×10^{-5} m² stretches uniformly by the same amount as another wire B of length 6.0 m and a cross section of 3.0×10^{-5} m² stretches. The ratio of the Young's modulus of wire A to that of wire B will be:

(1) 1 : 4 (2) 1 : 1
(3) 1 : 10 (4) 1 : 2

Space for your notes:

Q9 - 01 February - Shift 1

A certain pressure 'P' is applied to 1 litre of water and 2 litre of a liquid separately. Water gets compressed to 0.01% whereas the liquid gets compressed to 0.03%. The ratio of Bulk modulus of water to that of the liquid is $\frac{3}{x}$. The value of x is

Space for your notes:

Q10 - 01 February - Shift 2

The Young's modulus of a steel wire of length 6 m and cross-sectional area 3 mm^2, is 2×11^{11} N/m^2. The wire is suspended from its support on a given planet. A block of mass 4 kg is attached to the free end of the wire. The acceleration due to gravity on the planet is $\frac{1}{4}$ of its value on the earth. The elongation of wire is (Take g on the earth = 10 m/s^2):

(1) 1 cm (2) 1 mm
(3) 0.1 mm (4) 0.1 cm

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (4) **Q2** (3) **Q3** (5) **Q4** (1)

Q5 (4) **Q6** (60) **Q7** (4) **Q8** (2)

Q9 (1) **Q10** (3)

Q1 (4)

Elongation in wire $\delta = \frac{F\ell}{AY}$

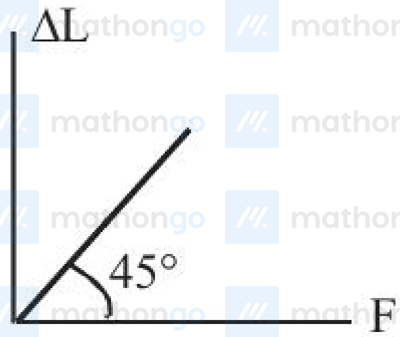
$$\delta = \frac{250 \times 100}{6.25 \times 10^{-4} \times 10^{10}}$$

$$\delta = 4 \times 10^{-3}\ \mathrm{m}$$

Q2 (3)

Concept based

Q3 (5)



From graph :

$$F = \Delta L$$

$$Y = \frac{FL}{A\Delta L}$$

$$Y = \frac{L}{A}$$

$$Y = \frac{62.8 \times 10^{-2}}{\pi\left(2 \times 10^{-3}\right)^2}$$

$$Y = 5 \times 10^4 \ N/m^2$$

Q4 (1)

$$Y = 3\eta(1+\sigma)$$

$$Y = 3K(1-\sigma)$$

$$\Rightarrow 2\eta(1+\sigma) = 3K(1-2\sigma)$$

$$\Rightarrow \sigma = \left(\frac{3K-2\eta}{6K+2\eta}\right)$$

Q5 (4)

$$Y = \frac{F/A}{\frac{\Delta \ell}{\ell}}$$

$$\Rightarrow \quad F = \frac{YA}{\ell}\Delta \ell$$

$$\left(\frac{A\Delta \ell}{\ell}\right)_1 = \left(\frac{A\Delta \ell}{\ell}\right)_2$$

$$\Rightarrow \quad \frac{\Delta \ell_2}{\Delta \ell_1} = \frac{A_1}{A_2} \times \frac{\ell_2}{\ell_1}$$

$$\Rightarrow \quad \frac{\Delta \ell_2}{0.2} = \frac{1}{2.4 \times 2.4} \times \frac{2}{1}$$

$$\Rightarrow \quad \Delta \ell_2 = 6.9 \times 10^{-2} \text{ mm}$$

Q6 (60)

If $\Delta\ell$ is decease in length of rod due to decease in temperature

$$\Delta\ell = \ell\alpha\Delta T$$

$$\alpha = 2\times10^{-5}K^{-1}, \Delta T = (210-160) = 50\ K$$

$$\Delta\ell = 1\times2\times10^{-5}\times50 = 10^{-3}\,m$$

$$\text{Young Modulus} = Y = \frac{F/A}{\Delta\ell/\ell} \quad A = 3\times10^{-6}\,m^2$$

$$2\times10^{11} = \frac{Mg/3\times10^{-6}}{10^{-3}/1}$$

$$Mg = 2\times10^{11}\times3\times10^{-9} = 6\times10^{-2}$$

$$M = 60\ kg$$

Ans is 60.

Q7 (4)

No Solution

Q8 (2)

No Solution

Q9 (1)

$$B_{water} = \frac{-\Delta P}{\left(\frac{\Delta V}{V}\right)} = \frac{-\Delta P}{\frac{0.01}{100}}$$

$$B_{liquid} = \frac{-\Delta P}{\frac{0.03}{100}}$$

$$\frac{B_{water}}{B_{liquid}} = 3$$

$x = 1$

Q10 (3)

$$\text{Tension (F)} = \text{mg}$$

$$= 4 \times \frac{10}{4} = 10\text{N}$$

$$\Delta\text{L} = \frac{\text{FL}}{\text{AY}}$$

$$= \frac{10 \times 6}{3 \times 10^{-6} \times 2 \times 10^{11}}$$

$$= 10^{-4}\ \text{m} = 0.1\ \text{mm}$$